COMPLEX ENGINEERING PROBLEM (CEP)

Course Title	Computer System Architecture
Instructor	Dr. Shahid Ismail
Lab Engineer	Engr. Usama Shoukat
Course Code	EC-220
Semester	3rd (Fall 2025)
Entry	DE46CSE
Credit Hours	3(Theory)+1(Lab)

Introduction:

Multipliers are fundamental hardware units in digital systems, providing the ability to perform fast and efficient multiplication operations that are central to a wide range of computational tasks. Unlike addition or subtraction, multiplication involves multiple partial product generations and accumulations, making it more complex and often a critical determinant of system performance. Their importance lies in enabling high-speed arithmetic required in applications such as digital signal processing (e.g., filtering, Fourier transforms, and image compression), graphics and multimedia (scaling, rotation, and rendering), cryptography (modular arithmetic in RSA, ECC, and block ciphers), scientific and engineering computations (matrix operations, simulations, and numerical methods), machine learning and artificial intelligence (dot products, convolutions, and matrix multiplications), and control systems (PID controllers, robotics, and real-time sensor fusion). By implementing multiplication in hardware, systems achieve significant speedups, reduce energy consumption compared to software-based iterative methods, and support the increasingly compute-intensive workloads of modern computing.

CEP Statement:

Design a hardware co-processor for multiplication in an integer-only RISC-V core (RV32I). You have to perform the following functions

1. Design a hardware multiplier co-processor and verify its functionality
2. Integrate the designed multiplier into the already built system
3. The multiplier must interface cleanly with the existing system.

Course	WK	PLO (WA)	WP	Bloom's Learning Level
Computer System Architecture	WK3, WK4, WK5, WK6	3 (Design)	WP1, WP2, WP3, WP4	C6

Deliverables:

1. Complete hardware structure of the multiplier considered
2. Project design report addressing all requirements
3. Complete System Verilog code of the project
4. Example instruction set for demonstration

Pre-Requisite:

To carry out this complex engineering problem you should have the knowledge and understanding of

1. Instruction set architecture
2. Integration of I and M extensions in RISCV
3. Hazards in computers
